2025 VCE Mathematical Methods 1 (NHT) external assessment report

This report provides sample answers or an indication of what answers may have included. Unless otherwise stated, these are not intended to be exemplary or complete responses.

Question 1a

This question involved using either the quotient rule or product rule for differentiation. The answer was required to be presented in a simplified form.

$$\frac{dy}{dx} = \frac{\frac{1}{x} \cdot x^3 - \log_e(x) \cdot 3x^2}{\left(x^3\right)^2}$$
$$= \frac{x^2 - 3x^2 \log_e(x)}{x^6}$$
$$= \frac{1 - 3\log_e(x)}{x^4} \quad \text{or} \quad x^{-4}(1 - 3\log_e(x))$$

Or using product rule

$$\frac{d}{dx}(\log_e(x).x^{-3})$$
$$\frac{dy}{dx} = \frac{1}{x} \cdot x^{-3} - \log_e(x) \cdot 3x^{-4}$$
$$= \frac{1}{x^4} - \frac{3\log_e(x)}{x^4} = \frac{1 - 3\log_e(x)}{x^4} \quad \text{or} \quad x^{-4}(1 - 3\log_e(x))$$

Question 1b

This question involved using the chain rule for differentiation and then evaluating the derivative at $x = 0$.

$$f'(x) = 3\pi e^{3x} \cdot \cos\left(\pi e^{3x}\right)$$

$$f'(0) = 3\pi e^0 \cos\left(\pi e^0\right)$$
$$= 3\pi \cos(\pi)$$
$$= -3\pi$$

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Question 2

The value of c is obtained either by observation of the y-intercept or by evaluation of $g(0)$. In order to find a and b, students needed to form two equations using the point (6, 9), with one involving the derivative.

$g(0) = 9 \Rightarrow c = 9$ from y-intercept

$$\begin{aligned} g(6) = 9 &\Rightarrow 216a + 36b + 9 = 9 \\ &\Rightarrow 216a + 36b = 0 \\ &\Rightarrow b = -6a \qquad (1) \end{aligned}$$

$g'(x) = 3ax^2 + 2bx$ and $h'(x) = \dfrac{x}{2}$

$$\begin{aligned} g'(6) &= h'(6) \\ &\Rightarrow 108a + 12b = 3 \\ &\Rightarrow 36a + 4b = 1 \qquad (2) \end{aligned}$$

Sub (1) into (2)

$$\begin{aligned} 36a + 4b = 1 \Rightarrow 12a = 1 &\Rightarrow a = \frac{1}{12} \\ \Rightarrow b = -6a &\Rightarrow b = -\frac{1}{2} \end{aligned}$$

Question 3a

This question required general solutions to be found. The general solution needed to include multiples of π. It is expected that students will have a way of remembering the exact values of $\sin\theta$, $\cos\theta$ and $\tan\theta$ for values of θ between 0 and $\dfrac{\pi}{2}$ inclusive, as specified in the key knowledge of the study design.

$$f(x) = 0 \Rightarrow \frac{1}{2}\tan(x) = -\frac{1}{2} \Rightarrow \tan(x) = -1$$

reference angle $x = -\dfrac{\pi}{4}$

$$x = -\frac{\pi}{4} + n\pi, \quad n \in Z, \; x = \frac{\pi}{4}(-1 + 4n\pi), \quad n \in Z$$

$$x = \frac{3\pi}{4} + n\pi, \quad n \in Z, \; x = \frac{\pi}{4}(3 + 4n), \quad n \in Z$$

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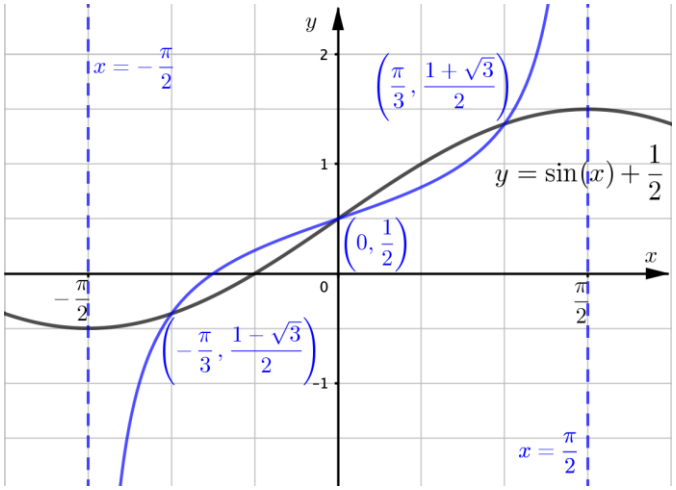
Question 3b.i

There were multiple ways in which this question could be attempted.

$$f(x) = g(x) \Rightarrow \frac{1}{2}\tan(x) + \frac{1}{2} = \sin(x) + \frac{1}{2}$$
$$\Rightarrow \frac{\sin(x)}{2\cos(x)} = \sin(x)$$
$$\Rightarrow \sin(x) = \sin(x)2\cos(x)$$
$$\Rightarrow \sin(x) - \sin(x)2\cos(x) = 0$$
$$\Rightarrow \sin(x)\left(1 - 2\cos(x)\right) = 0$$
$$\Rightarrow \sin(x) = 0 \quad \text{or} \quad \cos(x) = \frac{1}{2}$$
$$\therefore \quad x = 0 \quad \text{or} \quad x = \frac{\pi}{3}, -\frac{\pi}{3}$$

Question 3b.ii

Graph lines should be smooth, single-stroke constructions and display appropriate asymptotic behaviour. Asymptotes should be marked in as dashed or dotted lines and labelled with their equation.



Question 4a

This question could be attempted by setting up a binomial probability distribution or by using a tree diagram to determine the combinations.

Let X be the number of bruised apples from four apples

$$X \sim \text{Bi}\left(n = 4,\ p = \frac{1}{3}\right)$$

$$\begin{aligned}\Pr(X = 3) &= \binom{4}{3}\left(\frac{1}{3}\right)^3\left(\frac{2}{3}\right)^1 \\ &= 4 \times \frac{1}{27} \times \frac{2}{3} \\ &= \frac{8}{81}\end{aligned}$$

Question 4b

This question involved finding the sum of probabilities. We can define the random variable $\hat{P} = \frac{Y}{10}$, where Y is a binomial random variable with parameters $n = 10, p = \frac{1}{3}$ and $\Pr\left(\hat{P} = p\right) = \Pr(Y = y)$.

The answer was required to be given in a particular format.

$$\text{Let } Y \sim \text{Bi}\left(n = 10, p = \frac{1}{3}\right)$$

$$\begin{aligned}\Pr\left(\hat{P} \le \frac{1}{10}\right) &= \Pr(0 \le Y \le 1) \\ &= \Pr(Y = 0) + \Pr(Y = 1) \\ &= \binom{10}{0}\left(\frac{1}{3}\right)^0\left(\frac{2}{3}\right)^{10} + \binom{10}{1}\left(\frac{1}{3}\right)^1\left(\frac{2}{3}\right)^9 \\ &= \frac{2^{10}}{3^{10}} + 10 \times \frac{1}{3} \times \frac{2^9}{3^9} = \frac{2 \times 2^9 + 10 \times 2^9}{3^{10}} \\ &= \frac{2^9 \times (2 + 10)}{3^{10}} = \frac{2^9 \times 12}{3^{10}} \\ &= \frac{2^9 \times 2^2 \times 3}{3^{10}} \\ &= \frac{2^{11}}{3^9}\end{aligned}$$

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Question 5a

This question required the solution of a quadratic equation involving exponential terms.

$$f(x) = g(x) \Rightarrow e^{2x} = e^x + 2$$
$$\left(e^x\right)^2 - \left(e^x\right) - 2 = 0$$
$$\text{let } u = e^x$$
$$u^2 - u - 2 = 0$$
$$(u-2)(u+1) = 0$$
$$u = 2, \quad u = -1$$
$$\therefore e^x = 2, \quad \cancel{e^x = -1} \text{ (rejected, as } e^x > 0)$$
$$\therefore x = \log_e(2) \quad \text{only}$$

Question 5b

This question involved equating the derivatives of the functions $f(x)$ and $g(x)$. There were multiple ways that the equation could be solved.

$$f'(x) = 2e^{2x}, \quad g'(x-1) = e^{x-1}$$
$$2e^{2x} = e^{x-1} \Rightarrow 2e^{2x} - e^{x-1} = 0, \quad 2e^x \cdot e^x - e^x.e^{-1} = 0$$
$$e^x\left(2e^x - e^{-1}\right) = 0$$
$$\cancel{e^x = 0} \text{ or } (2e^x - e^{-1}) = 0$$
$$\Rightarrow e^x = \frac{1}{2e}$$
$$\Rightarrow x = \log_e\left(\frac{1}{2e}\right) \text{ or } x = -\log_e(2e)$$
$$\left(\begin{array}{l}\text{or } x = -\log_e(2) - 1 \\ \text{or other equivalent forms}\end{array}\right)$$

Alternatively

$$2e^{2x} = e^{x-1} = \frac{e^x}{e}, \quad e^x > 0$$
$$\Rightarrow 2e^x = \frac{1}{e}$$
$$\therefore e^x = \frac{1}{2e} \Rightarrow x = \log_e\left(\frac{1}{2e}\right)$$

Alternatively

$$2e^{2x} = e^{x-1}$$
Then taking natural log both sides
$$\log_e\left(2e^{2x}\right) = \log_e\left(e^{x-1}\right)$$
$$\log_e(2) + \log_e(e^{2x}) = (x-1)\cancel{\log_e(e)}$$
$$\log_e(2) + 2x = x - 1$$
$$x = -1 - \log_e(2)$$

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Question 6a.i

n	1	2	3
$\Pr(N=n)$	0.6	0.2	0.2

Question 6a.ii

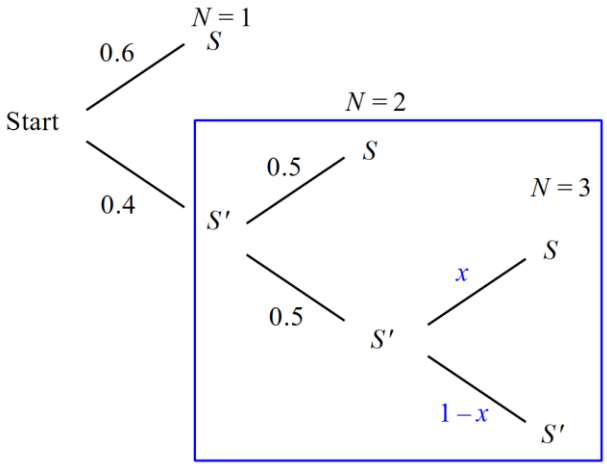
$$E(N)=\frac{8}{5}=1.6$$

Question 6b

A tree diagram was useful. Familiarity with probability notation, particularly conditional probability notation, was important.

$$\Pr(B\mid A')+\Pr(B'\mid A')\times\Pr(C\mid B')=0.7$$
$$0.5+0.5\Pr(C\mid B')=0.7$$
$$0.5\Pr(C\mid B')=0.2$$
$$\therefore\quad \Pr(C\mid B')=0.4$$

$$0.5x+0.5=0.7$$
$$0.5x=0.2$$
$$\therefore\ x=0.4$$
$$\therefore\quad \Pr(C\mid B')=0.4$$



Alternatively

$$0.5(1-x)=0.3$$
$$1-x=0.6$$
$$\therefore\ x=0.4$$
$$\therefore\quad \Pr(C\mid B')=0.4$$

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Question 7a

There were multiple methods that students could use for this question.

Method 1: Using quadratic formula

$$x = -\frac{b}{2} \pm \frac{\sqrt{b^2+24}}{2}$$
$$\Delta x = x_2 - x_1 = 5$$
$$\Delta x = -\frac{b}{2} + \frac{\sqrt{b^2+24}}{2} - \left(-\frac{b}{2} - \frac{\sqrt{b^2+24}}{2}\right) = 2 \times \frac{\sqrt{b^2+24}}{2}$$
$$= \sqrt{b^2+24} = 5$$
$$\Rightarrow b^2 + 24 = 25, \quad b^2 = 1$$
$$\Rightarrow b = \pm 1$$

Method 2: Using completing the square

$$x^2 + bx - 6 = 0 \Rightarrow x^2 + bx + \frac{b^2}{4} - \frac{b^2}{4} - 6 = 0$$
$$\left(x + \frac{b}{2}\right)^2 - \left(\frac{24 + b^2}{4}\right) = 0$$
$$\left(x + \frac{b}{2}\right)^2 = \left(\frac{24 + b^2}{4}\right)$$
$$x + \frac{b}{2} = \pm \frac{\sqrt{b^2+24}}{2}$$
$$\therefore x = -\frac{b}{2} \pm \frac{\sqrt{b^2+24}}{2}$$
$$\Delta x = x_2 - x_1 = 5$$
$$\Delta x = -\frac{b}{2} + \frac{\sqrt{b^2+24}}{2} - \left(-\frac{b}{2} - \frac{\sqrt{b^2+24}}{2}\right)$$
$$= \sqrt{b^2+24} = 5$$
$$\Rightarrow b^2 + 24 = 25, \quad b^2 = 1$$
$$\Rightarrow b = \pm 1$$

Method 3: Using roots

Let the 2 roots be a and c

$ac = -6 \quad \text{eqn}(1)$

$a - c = 5 \quad \text{eqn}(2)$

$\therefore c(c+5) = -6$

$c^2 + 5c + 6 = 0$

$c = -2, c = -3$

$a = 3, a = 2$

so $(x+3)(x-2)$ or $(x+2)(x-3)$

$x^2 + x - 6$ or $x^2 - x - 6$

$\therefore b = 1$ or $b = -1$

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Question 7b.i

This question required students to explain why the particular function f does not have an inverse function when restricted to the domain $(0, \infty)$. The explanation needed to include the fact that the turning point lies in the interval $(0, \infty)$ and thus the function is not one-to-one (or is many-to-one).

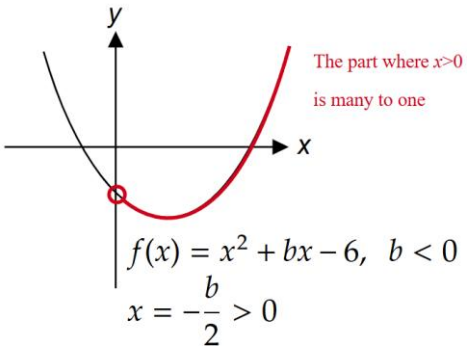
Local min at $x = -\frac{b}{2}$

$x_{sp} = -\frac{b}{2} > 0 \quad (\text{as } b < 0)$

Thus, $f(x)$ will be many-to-one/not one-to-one,
so f^{-1} isn't defined.

Alternatively, a diagram may be utilised to explain that $f(x)$ is many-to-one over $x \in (0, \infty)$, and therefore the inverse does not exist.

For example



Question 7b.ii

There were multiple methods that could be employed to find the inverse function. It was important to note that the domain was restricted to $(-\infty, 0)$, as this affected the choice of the branch of the square root in the inverse function. Common methods included completing the square or using the quadratic formula.

Method 1: Completing the square

For inverse let

$$x = y^2 + by - 6$$
$$= \left(y + \frac{b}{2}\right)^2 - \frac{b^2 + 24}{4}$$

$$x + \frac{b^2 + 24}{4} = \left(y + \frac{b}{2}\right)^2$$
$$\pm\sqrt{x + \frac{b^2 + 24}{4}} = y + \frac{b}{2}$$
$$\text{As } \text{dom}(f) = (-\infty, 0) = \text{range}\left(f^{-1}\right)$$
$$y = f^{-1}(x) < 0$$
$$\therefore \quad f^{-1}(x) = -\frac{b}{2} - \sqrt{x + \frac{b^2 + 24}{4}}$$

$$\text{where } x > -6 \quad \left(x \in (-6, \infty)\right)$$

Method 2: Using quadratic formula

$$x = y^2 + by - 6$$
$$0 = y^2 + by - (x + 6)$$
$$\therefore y = \frac{-b}{2} \pm \frac{\sqrt{b^2 + 4(x+6)}}{2}$$
$$y = \frac{-1}{2}\left(b + \sqrt{b^2 + 4(x+6)}\right), \text{ as } y < 0$$
$$f^{-1}(x) = \frac{-1}{2}\left(b + \sqrt{b^2 + 4(x+6)}\right) = \frac{-1}{2}\left(b + \sqrt{b^2 + 4x + 24}\right)$$

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Question 7c

There were multiple ways in which this question could be explored. For these types of questions, students are encouraged to sketch a graph to assist in developing their solutions.

$$\begin{aligned} g \circ f(x) &= \log_e\left(-f(x)-2\right) \\ &= \log_e\left(-x^2-bx+6-2\right) \\ &= \log_e\left(-x^2-bx+4\right) \end{aligned}$$

For $g \circ f(x)$ to be defined

$-x^2-bx+4>0$ over $(-2,2)$

$\therefore -x^2-bx+4=-(x+2)(x-2)=-x^2+4$

$\therefore b=0$

Alternatively

By inspection, if $b=0$, then $-x^2+4>0$

$\Rightarrow x^2-4<0$

$\Rightarrow x^2<4$

$\Rightarrow -2<x<2$

Hence $b=0$.

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Question 8a.i

This question was a ‘show that’ question. As such, each line of working needed to demonstrate a clear, logical and explicit progression, leading to the answer. In particular, the product rule needed to be clearly demonstrated.

Using product rule

$$\text{so } \frac{d}{dx}\left(x\sqrt{x+6}\right)=\frac{d}{dx}(x).\sqrt{x+6}+x.\frac{d}{dx}\left(\sqrt{x+6}\right)$$

$$=1.\sqrt{x+6}+x.\frac{1}{2}(x+6)^{\frac{-1}{2}}=\sqrt{x+6}+x.\frac{1}{2}(x+6)^{\frac{-1}{2}}$$

$$=1\cdot\sqrt{x+6}+x\cdot\frac{1}{2\sqrt{x+6}}$$

$$=\sqrt{x+6}+\frac{x}{2\sqrt{x+6}}$$

Alternatively

$$\text{let } u=x, \quad v=\sqrt{x+6}$$

$$\frac{du}{dx}=1, \quad \frac{dv}{dx}=\frac{1}{2\sqrt{x+6}}$$

$$\frac{d}{dx}(x\sqrt{x+6})=x.\frac{1}{2\sqrt{x+6}}+1.\sqrt{x+6}$$

$$=\frac{x}{2\sqrt{x+6}}+\sqrt{x+6}$$

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Question 8a.ii

The most common approach to evaluating the definite integral was to use the 'hence' option and set up the integration by recognition of the result in Question 8a.i.

$$k = 6 \Rightarrow f(x) = \frac{x}{2\sqrt{x+6}}$$

$$\frac{d}{dx}\left(x\sqrt{x+6}\right) = \sqrt{x+6} + f(x)$$

$$\text{so } f(x) = \frac{d}{dx}\left(x\sqrt{x+6}\right) - \sqrt{x+6}$$

$$\text{so } \int_0^6 f(x)dx = \int_0^6 \left(\frac{d}{dx}\left(x\sqrt{x+6}\right) - \sqrt{x+6}\right)dx$$

$$= \left[x\sqrt{x+6} - \frac{2}{3}(x+6)^{\frac{3}{2}}\right]_0^6$$

$$= \left(6\sqrt{12} - \frac{2}{3}(12)^{\frac{3}{2}}\right) - \left(0 - \frac{2}{3}(6)^{\frac{3}{2}}\right)$$

$$= 12\sqrt{3} - \frac{2}{3}\cdot 12\cdot 2\sqrt{3} + \frac{2}{3}\cdot 6\cdot\sqrt{2}\cdot\sqrt{3}$$

$$= -4\sqrt{3} + 4\sqrt{2}\cdot\sqrt{3}$$

$$= 4\sqrt{3}\left(\sqrt{2} - 1\right)$$

Question 8b

This question was a general version of the integral in Question 8a.ii.

From Question 8a.i:

$$\frac{d}{dx}\left(x\sqrt{x+k}\right) = \sqrt{x+k} + \frac{x}{2\sqrt{x+k}}$$

$$= \sqrt{x+k} + f(x)$$

$$\int_0^k f(x)dx = \int_0^k \left(\frac{d}{dx}\left(x\sqrt{x+k}\right) - \sqrt{x+k}\right)dx$$

$$= \left[x\sqrt{x+k} - \frac{2}{3}(x+k)^{\frac{3}{2}}\right]_0^k$$

$$= k\sqrt{2k} - \frac{2}{3}(2k)^{\frac{3}{2}} - \left(0 - \frac{2}{3}k^{\frac{3}{2}}\right)$$

$$= k\sqrt{k}\sqrt{2} - \frac{2}{3}\cdot 2\sqrt{2}\cdot k\sqrt{k} + \frac{2}{3}k\sqrt{k}$$

$$= k\sqrt{k}\left(\frac{3}{3}\sqrt{2} - \frac{4}{3}\sqrt{2} + \frac{2}{3}\right)$$

$$= \left(\frac{2-\sqrt{2}}{3}\right)k\sqrt{k}$$

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